

OBJECT ORIENTED PROGRAM

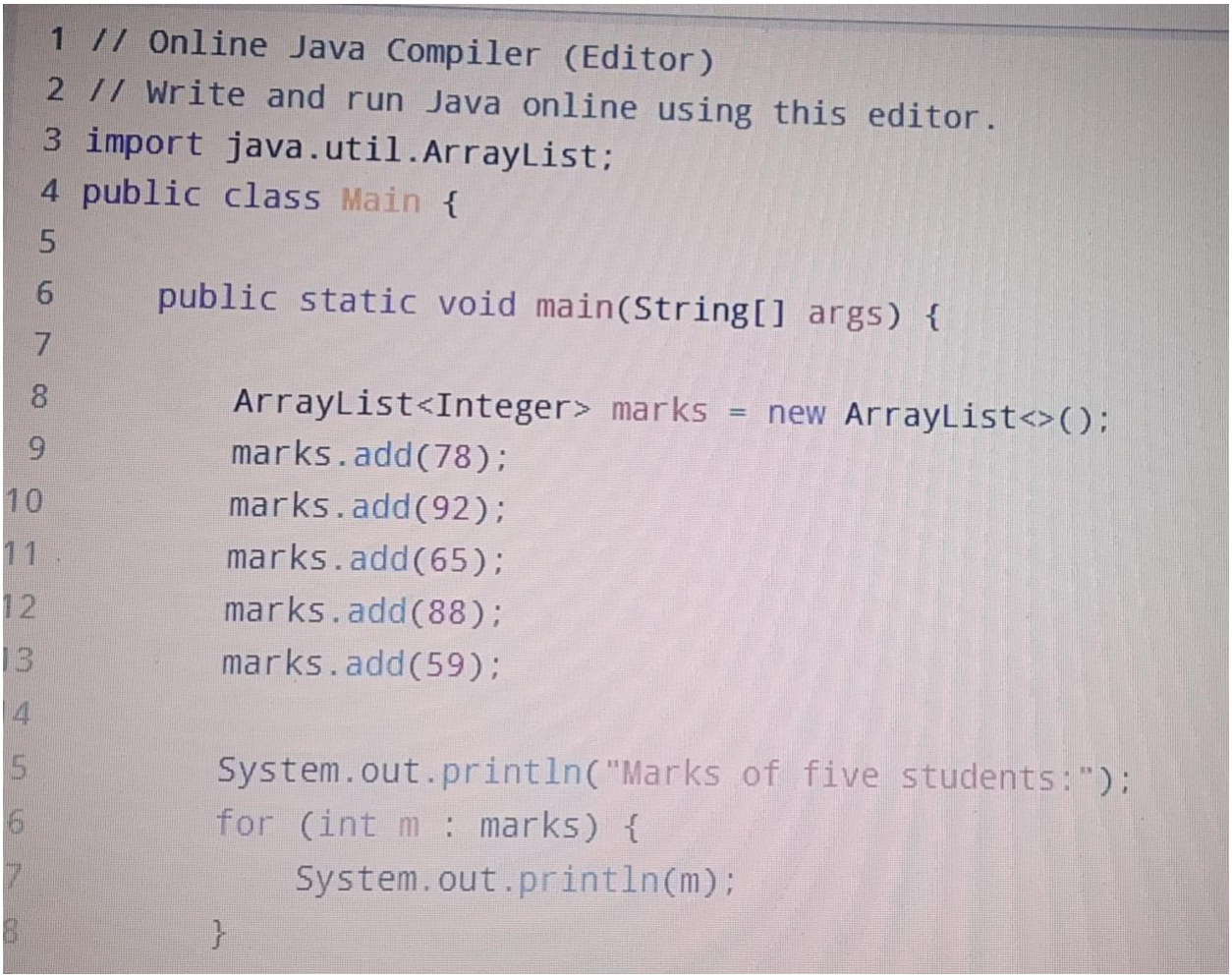
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Task 1

Write a Java program to store the marks of five students using an Array List<Integer>. Use an Enhanced For Loop to display all the marks, calculate and display the total and average marks, and find and display the highest mark.



```java
// Online Java Compiler (Editor)
// Write and run Java online using this editor.
import java.util.ArrayList;
public class Main {

    public static void main(String[] args) {

        ArrayList<Integer> marks = new ArrayList<>();
        marks.add(78);
        marks.add(92);
        marks.add(65);
        marks.add(88);
        marks.add(59);

        System.out.println("Marks of five students:");
        for (int m : marks) {
            System.out.println(m);
        }
```

```
17              System.out.println(m);
18          }
19
20          int total = 0;
21          for (int m : marks) {
22              total += m;
23          }
24          double average = (double) total / marks.size();
25
26          int highest = marks.get(0);
27          for (int m : marks) {
28              if (m > highest) {
29                  highest = m;
30              }
31          }
32
33          System.out.println("Total Marks: " + total);
34          System.out.println("Average Marks: " + average);
35          System.out.println("Highest Mark: " + highest);
```

Online Java Compiler

Main.java

Run

```
23          }
24          double average = (double) total / marks.size();
25
26          int highest = marks.get(0);
27          for (int m : marks) {
28              if (m > highest) {
29                  highest = m;
30              }
31          }
32
33          System.out.println("Total Marks: " + total);
34          System.out.println("Average Marks: " + average);
35          System.out.println("Highest Mark: " + highest);
36      }
37  }
38
```

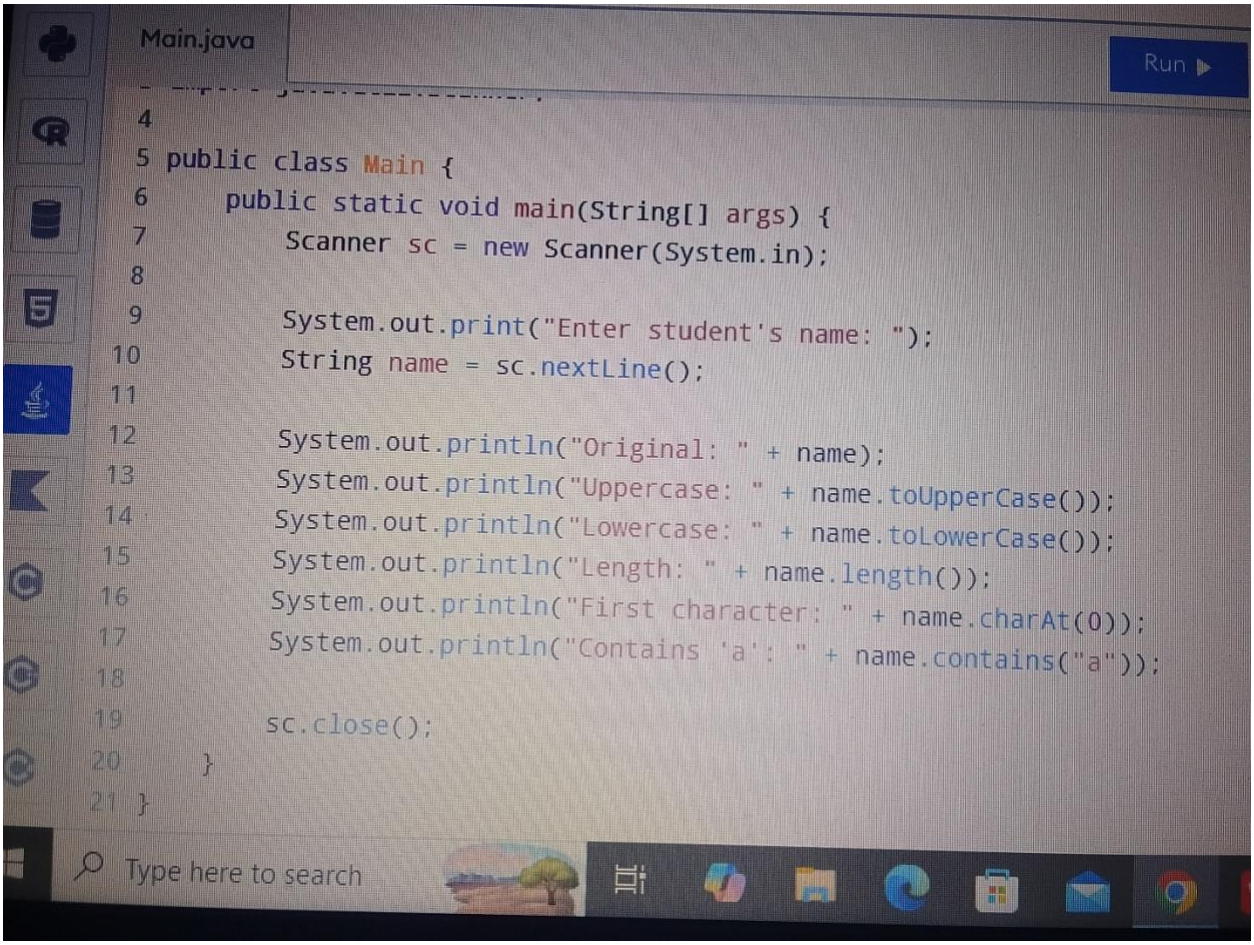
Output:

Output

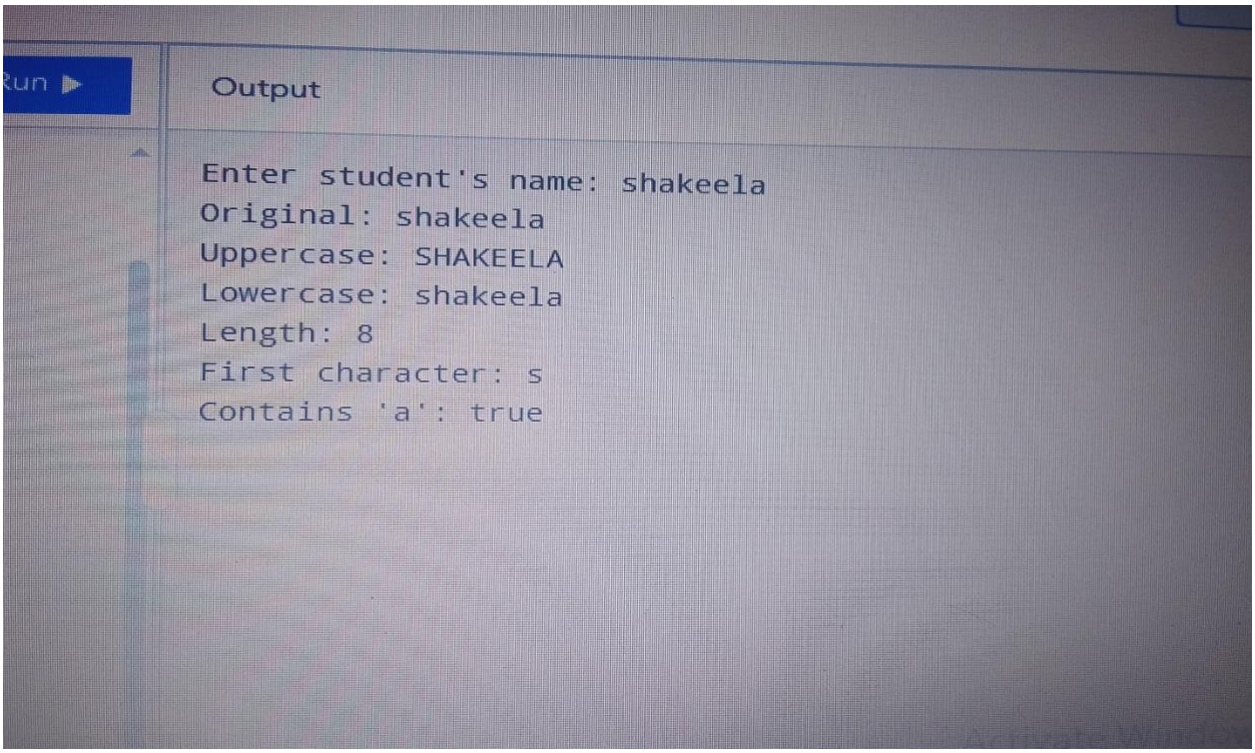
```
Marks of five students:
78
92
65
88
59
Total Marks: 382
Average Marks: 76.4
Highest Mark: 92
```

Task 2

Write a Java program that takes a student's name as a String and performs basic string processing. The program should display the original name, its uppercase and lowercase forms, the total number of characters, the first character using charAt(), and whether the name contains the letter "a"

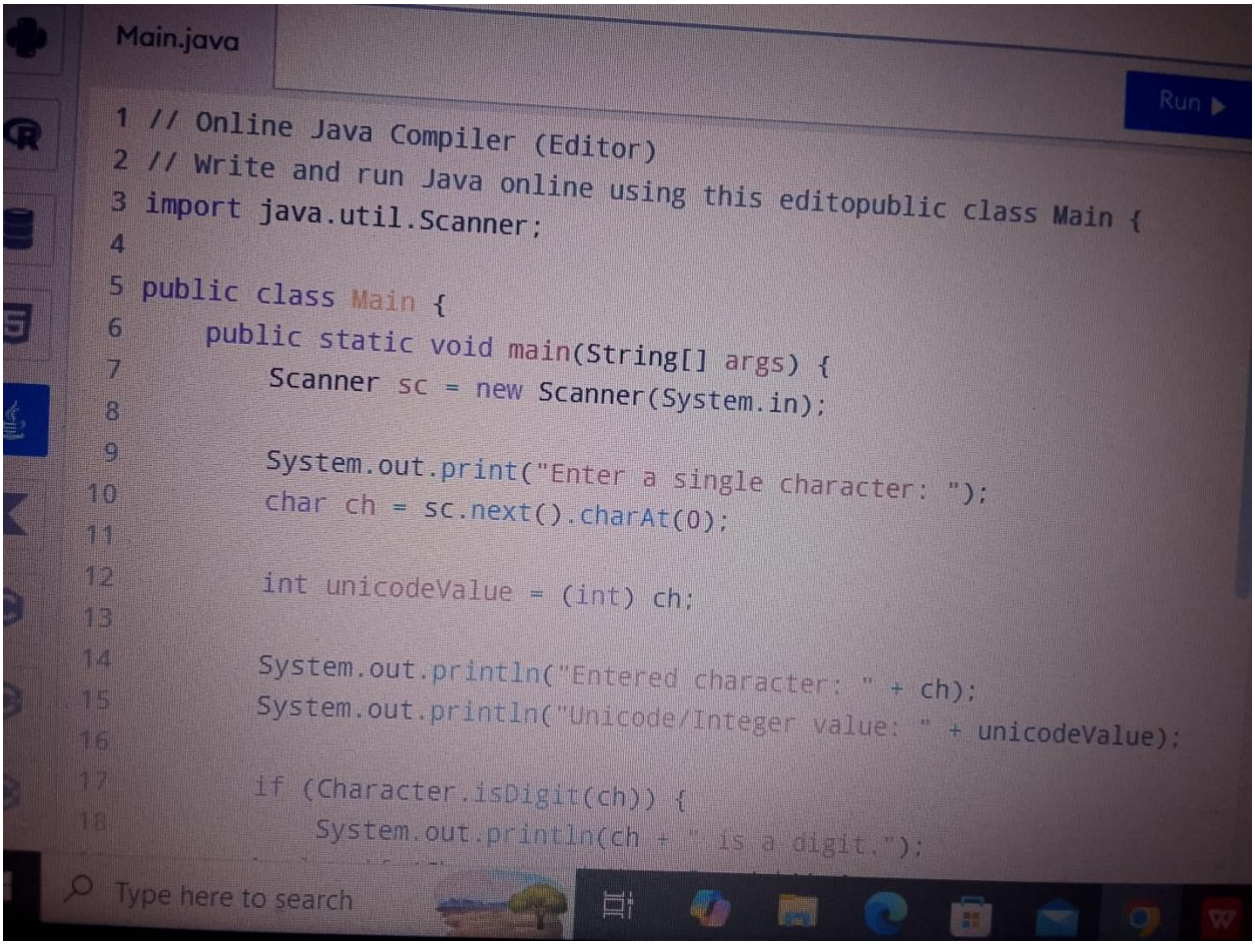


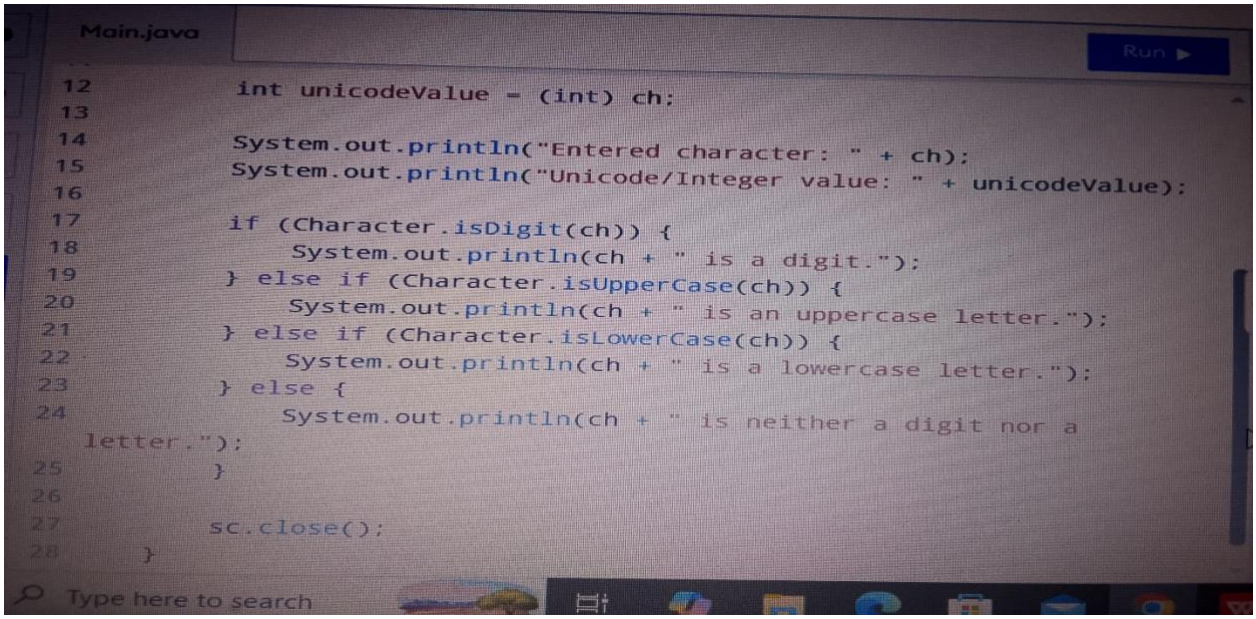
Output:



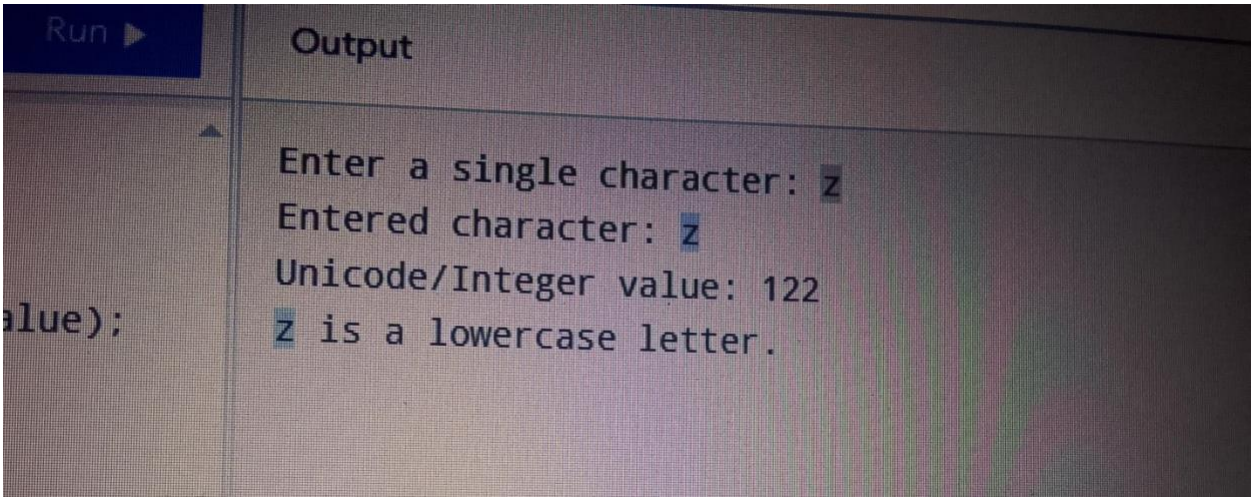
Task 3

Write a Java program that takes a single character from the user and stores it in a char variable. Display the entered character and its corresponding integer/Unicode value. The program should also determine whether the character is a digit, an uppercase letter, or a lowercase letter, and display the appropriate message.


Main.java
Run
1 // Online Java Compiler (Editor)
2 // Write and run Java online using this editopublic class Main {
3 import java.util.Scanner;
4
5 public class Main {
6 public static void main(String[] args) {
7 Scanner sc = new Scanner(System.in);
8
9 System.out.print("Enter a single character: ");
10 char ch = sc.next().charAt(0);
11
12 int unicodeValue = (int) ch;
13
14 System.out.println("Entered character: " + ch);
15 System.out.println("Unicode/Integer value: " + unicodeValue);
16
17 if (Character.isDigit(ch)) {
18 System.out.println(ch + " is a digit.");
Type here to search


Main.java
Run
12 int unicodeValue = (int) ch;
13
14 System.out.println("Entered character: " + ch);
15 System.out.println("Unicode/Integer value: " + unicodeValue);
16
17 if (Character.isDigit(ch)) {
18 System.out.println(ch + " is a digit.");
19 } else if (Character.isUpperCase(ch)) {
20 System.out.println(ch + " is an uppercase letter.");
21 } else if (Character.isLowerCase(ch)) {
22 System.out.println(ch + " is a lowercase letter.");
23 } else {
24 System.out.println(ch + " is neither a digit nor a letter.");
25 }
26
27 sc.close();
28 }
Type here to search

Output:


Run
Output
alue);
Enter a single character: z
Entered character: z
Unicode/Integer value: 122
z is a lowercase letter.